Prabhavit Gupta

Data Scientist | Retrieval, Agentic Systems, Semantic Search & Risk Narratives

+39 351 610 7807 | prabhavitg@gmail.com | linkedin.com/in/prabhavit | github.com/prab-gupta | Venice, Italy | Unrestricted EU Permanent Resident | Open to Relocate to Madrid

Summary

Data Scientist and AI Systems Engineer with 3 years of commercial experience designing, tuning, and deploying production hybrid retrieval systems, RAG pipelines, and autonomous agentic workflows. Technical background complemented by an academic degree in International Studies and Economics from Ca' Foscari University of Venice, providing deep analytical interest in global trade, sanctions, corporate networks, and financial intelligence. Proven expertise optimizing semantic search through hybrid sparse/dense retrieval (BM25 + vector embeddings), reranking models, and graph-aware context synthesis for narrative generation. Unrestricted EU permanent resident, immediately available and enthusiastic about hybrid work in Madrid.

Experience

FlowSpace.co • *Lead AI Engineer & Retrieval Architect (Freelance)*
Mar 2024 - Aug 2026 • London, UK (remote)

- Architected and deployed production semantic search and retrieval-augmented generation (RAG) platforms serving 4,500+ active enterprise users analyzing complex relational and document data.
- Engineered hybrid retrieval pipelines combining BM25 keyword matching with dense vector embeddings (pgvector, Chroma, Qdrant) and cross-encoder reranking, increasing retrieval recall@k by 28%.
- Developed autonomous agentic workflows (LangGraph) that traverse multi-hop entity relationships and synthesize disparate analytical signals into coherent, structured risk intelligence summaries.
- Integrated knowledge graph entity relationships and sub-network metadata into search retrieval layers, enabling contextual entity resolution and network risk narrative generation.
- Implemented continuous retrieval evaluation frameworks (Hit Rate, MRR, NDCG, RAG Triad faithfulness) to objectively measure search relevance and mitigate generative hallucinations.
- *Technologies: Python, SQL, BM25, Vector Search, pgvector, Qdrant, Reranking, LangGraph, FastAPI, Docker, AWS, Git*

Urlink • *Data Scientist & Machine Learning Engineer (Freelance)*
Apr 2025 - Jul 2025 • Barcelona, Spain (remote)

- Built entity extraction and semantic indexing pipelines to process heterogeneous unstructured trade and commercial data streams into queryable formats.
- Implemented automated data validation schemas and similarity metrics to resolve entity duplicates and ambiguous network links.
- Optimized SQL query performance and vector index clustering for low-latency similarity search across high-volume collections.
- *Technologies: Python, Pandas, PostgreSQL, pgvector, Supabase, Sentence-Transformers, REST APIs, Git*

Independent Technical Consultant • *AI Solutions Engineer*
Sep 2023 - Mar 2024 • Venice, Italy

- Delivered custom NLP, semantic search, and document intelligence pipelines for European enterprise clients handling regulatory compliance and operational records.
- Evaluated embedding models, chunking strategies, and prompt chains to maximize extraction fidelity from complex legal and corporate texts.
- Authored technical architecture documents, benchmark reports, and deployment guidelines for production AI integrations.
- *Technologies: Python, SQL, OpenAI API, LangChain, LlamaIndex, Chroma, Linux, Git*

Tractionboard.co • *Full-Stack Automation Developer (Intern)*
Aug 2023 - Oct 2023 • Utrecht, Netherlands (remote)

- Developed automated data ingestion pipelines and API webhooks for SaaS analytics dashboards.
- *Technologies: Python, JavaScript, REST APIs, Git*

Education

Ca' Foscari University of Venice • *Bachelor's in Philosophy, International Studies and Economics (Minor in Entrepreneurship)*
Oct 2020 - Jun 2024 • Venice, Italy

- Rigorous coursework in international economics, global geopolitical dynamics, institutional governance, quantitative analysis, and decision theory.

Certifications

Microsoft: Introduction to building AI agents • Microsoft Copilot Studio Agent Academy • Microsoft Power Automate • Tray.AI Build Practitioner (Sep 2024) • Tray.AI Build Foundations

Additional Information

- **Work Authorization:** Permanent EU Resident with unrestricted working rights across the European Union (Spain included, no visa sponsorship required).
- **Languages:** English (Fluent C1 professional), Italian (Working proficiency), Hindi (Native).
- **Availability & Relocation:** Immediate availability (0 days notice); ready for hybrid 3-days/week on-site in Madrid.

Core Skills

- **Retrieval & Search Systems:** Semantic Search, BM25, Hybrid Search, Dense Embeddings, Reranking (Cross-Encoders), pgvector, Qdrant, Chroma, Vector Indexing.
- **Agentic Architectures & RAG:** Multi-Agent Orchestration, LangGraph, Tool Calling, Entity Resolution, Contextual Narrative Generation, Evals (MRR, NDCG).
- **Languages & Data Engineering:** Python 3.12, SQL (PostgreSQL), PyTorch, Scikit-learn, Pandas, FastAPI, Pydantic.
- **Domain & Infrastructure:** International Trade & Sanctions Risk Interest, Graph-Aware Synthesis, Docker, AWS, Git, CI/CD, Linux.